CPH 200C PS3 · CLINICAL AI BY REGION

Three Governing Systems for Clinical AI

Why deployed clinical AI looks different across the US, Europe, and Asia: policy, funding, technical developments, and landscape

Curtis Chambers · Jinchi Wei

CPH 200C · Computational Precision Health

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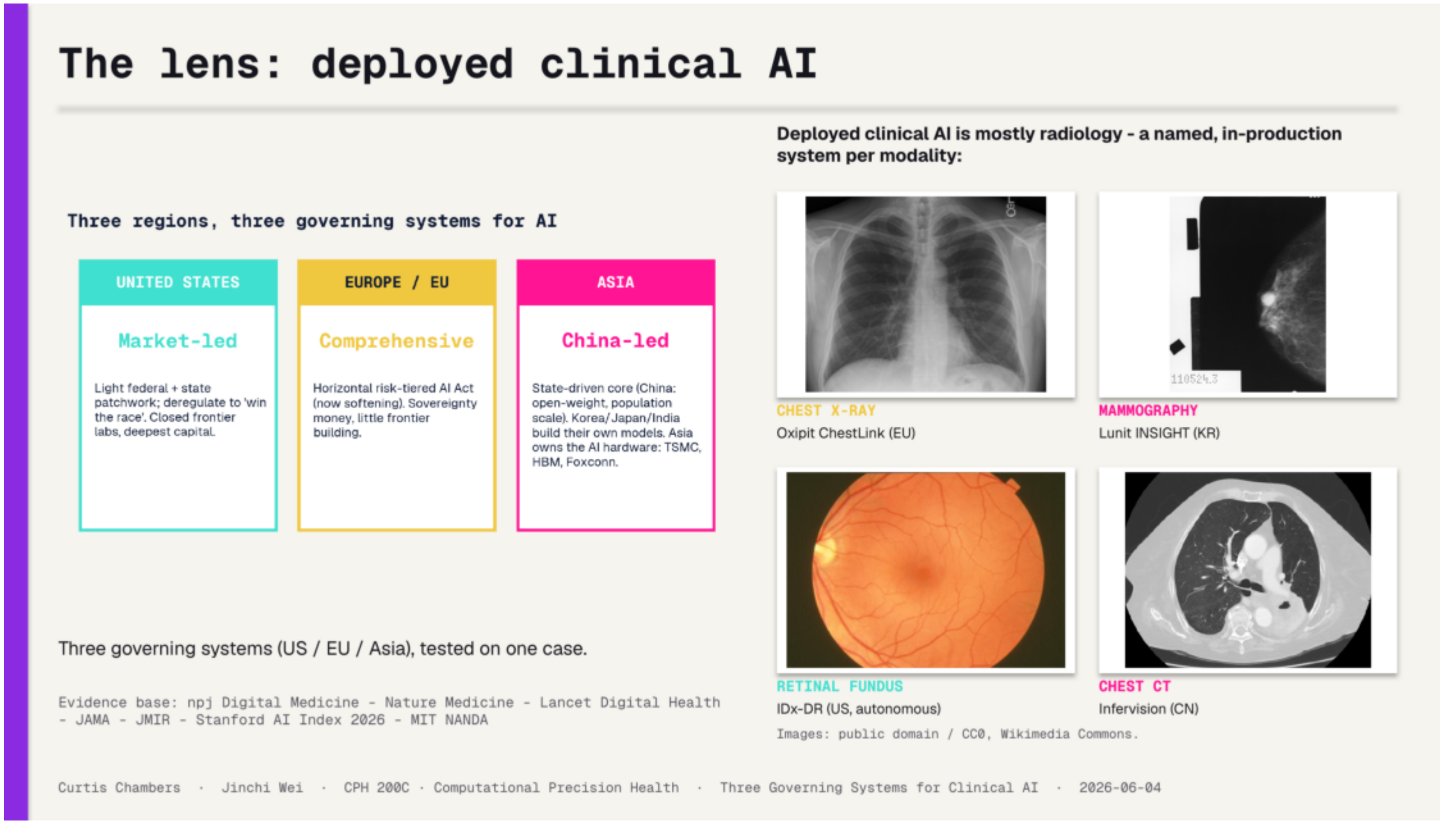
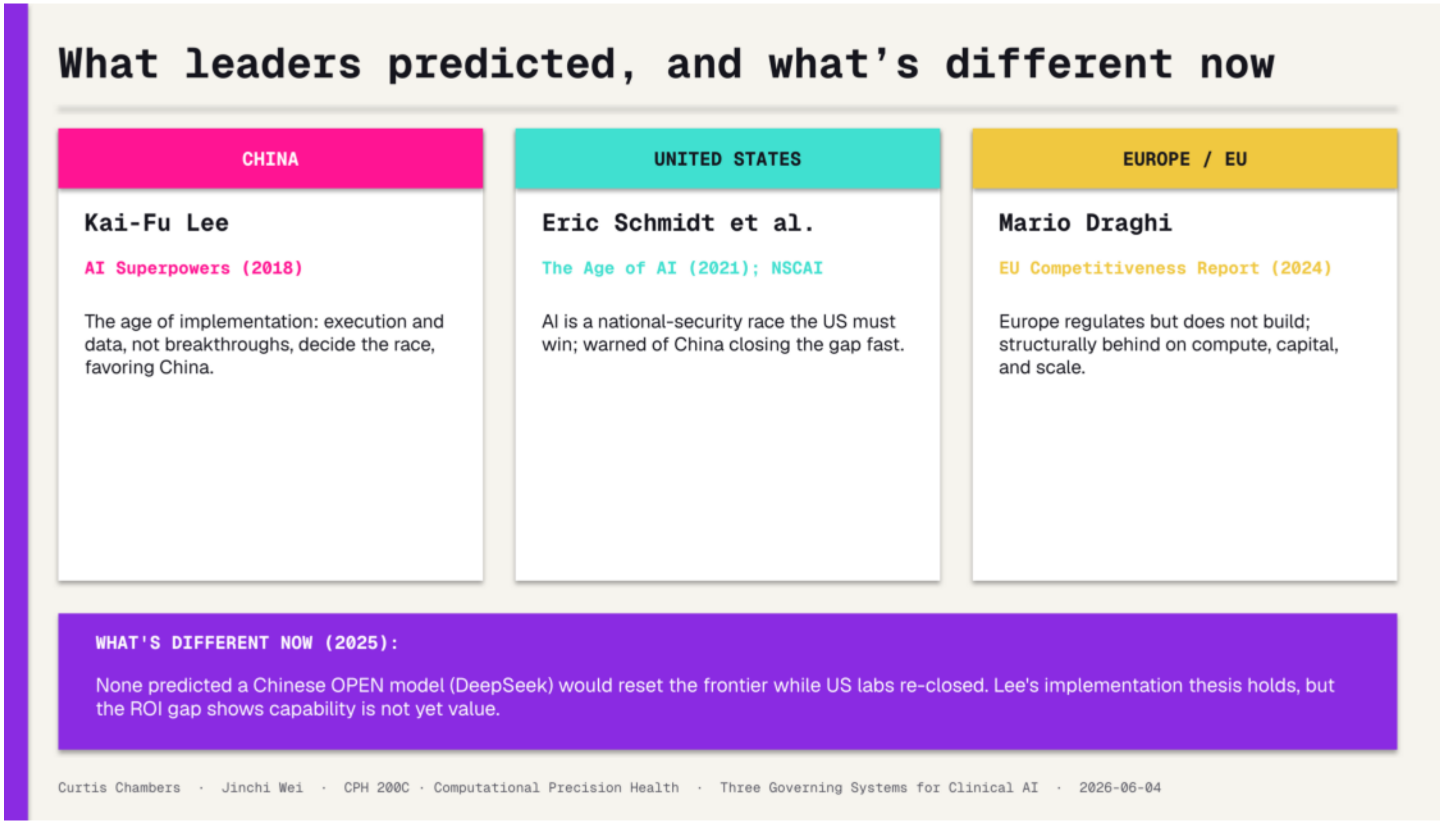
Slide 1 –



Slide 2 — The lens: deployed clinical AI

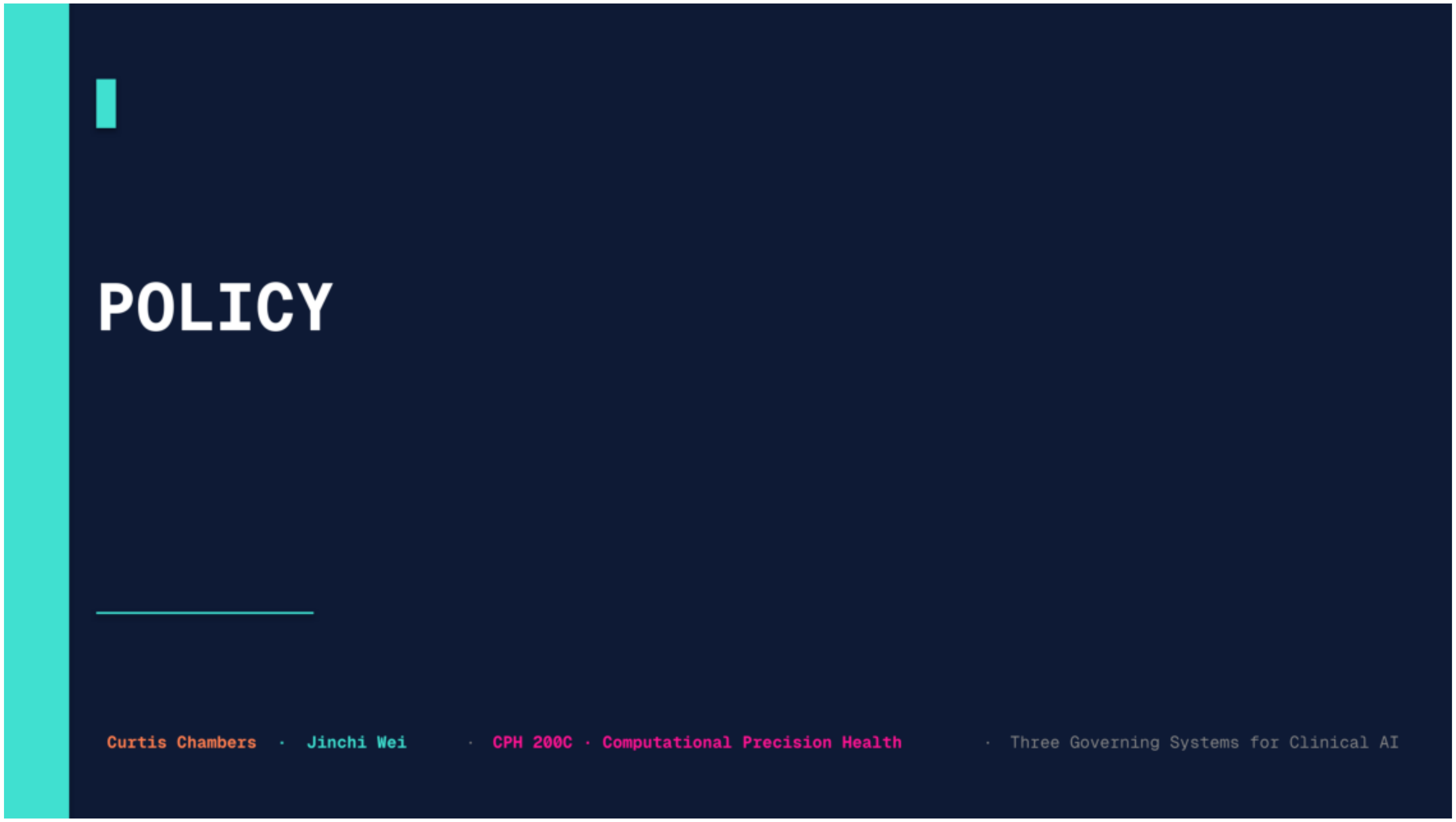
The lens

Set the frame: three regional AI governing systems, tested on deployed clinical AI because that is where the peer-reviewed evidence is strongest and the divergence sharpest. The radiology gallery makes it concrete: most deployed clinical AI is imaging, and each modality already has a named, in-production system, chest X-ray (Oxipit ChestLink, the EU autonomy milestone), mammography (Korea's Lunit INSIGHT, FDA-cleared), retinal fundus (IDx-DR, the first FDA-cleared fully autonomous diagnostic AI, in the US), and chest CT (China's Infervision, about 20,000 lung scans a day). Note the spread maps onto the regional thesis. Read the evidence-base line so the audience knows this is research-grounded. Then walk the four axes: policy, funding, technical, landscape.



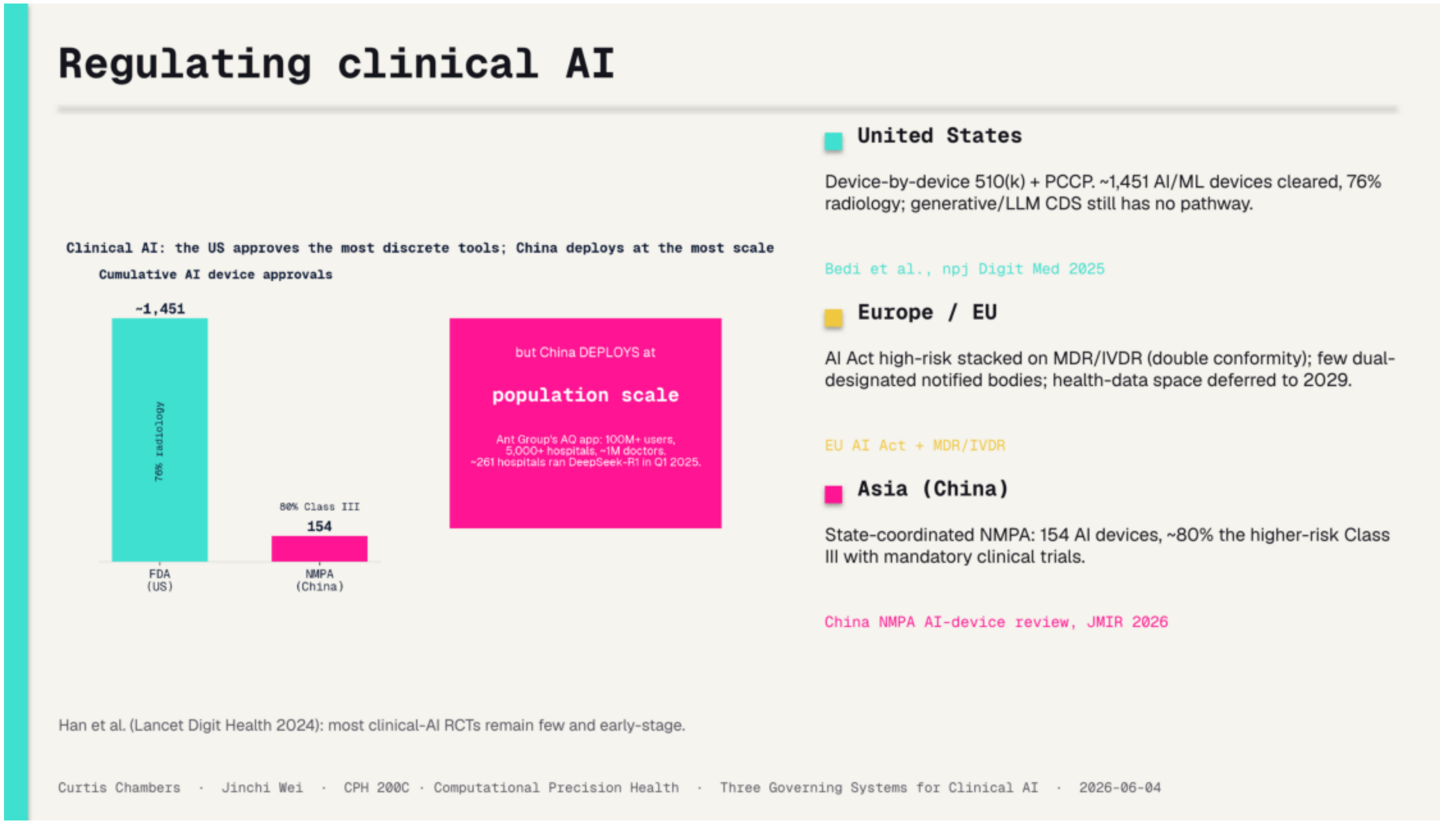
Slide 3 — What leaders predicted, and what's different now

Predictions vs now
This is the framing slide, and it earns the research framing. Three influential thinkers each forecast the AI race. China: Kai-Fu Lee's AI Superpowers (2018) argued we had entered the age of implementation, where execution and data, not breakthroughs, decide the winner, which favored China. US: Eric Schmidt, co-author of The Age of AI and chair of the National Security Commission on AI, framed it as a national-security race the US must win and warned China was closing fast. Europe: Mario Draghi's 2024 EU competitiveness report diagnosed that Europe regulates but does not build. What is different now, the purple band: the 2025 open-weight surge surprised everyone, a Chinese OPEN model reset the frontier while US labs re-closed, the opposite of what was expected. Lee's implementation thesis largely held, but the ROI gap shows capability is not yet value.
Transition: with that frame, the four axes, starting with policy.



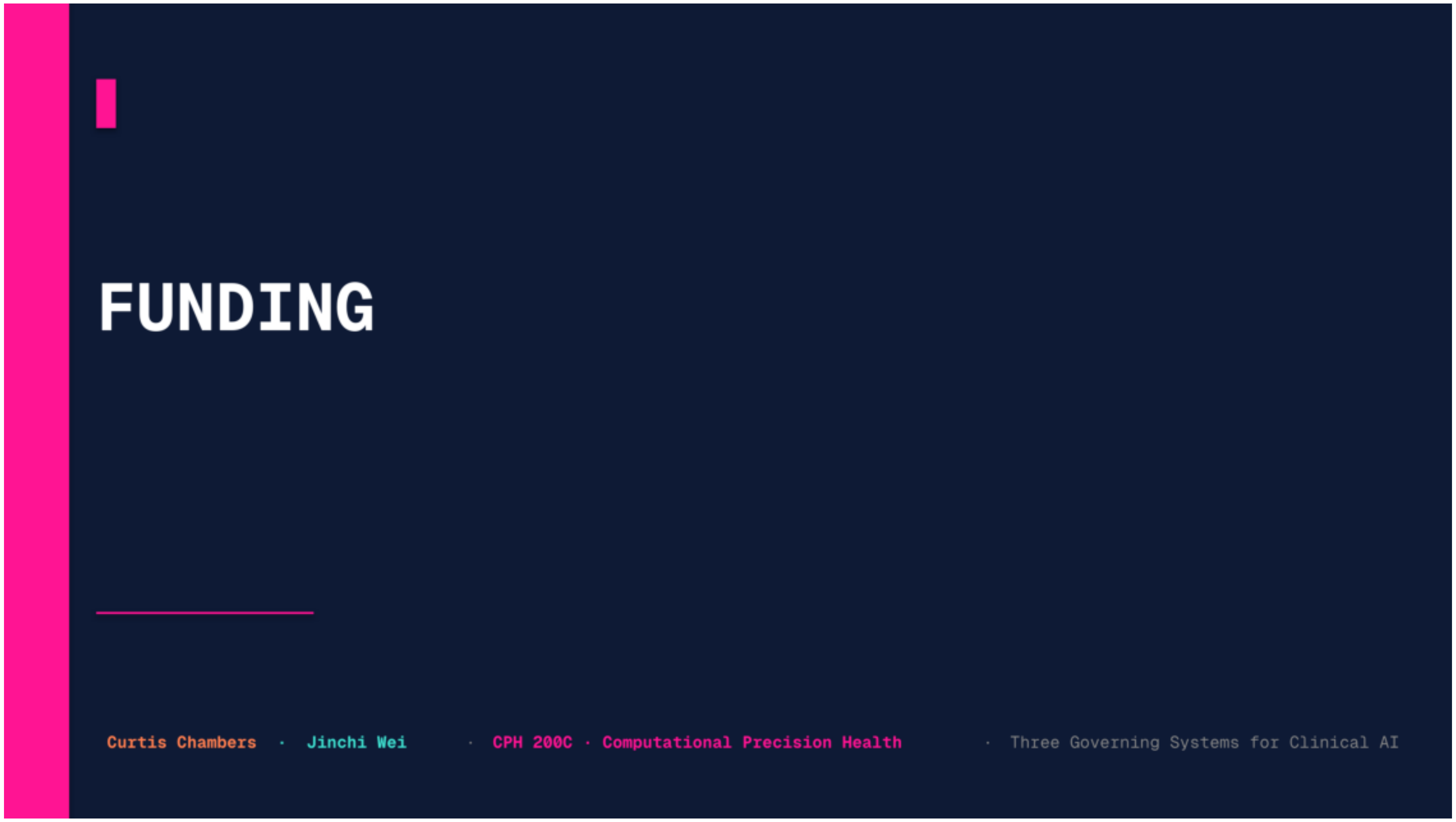
Slide 4 — Policy

Transition: how each region regulates clinical AI.



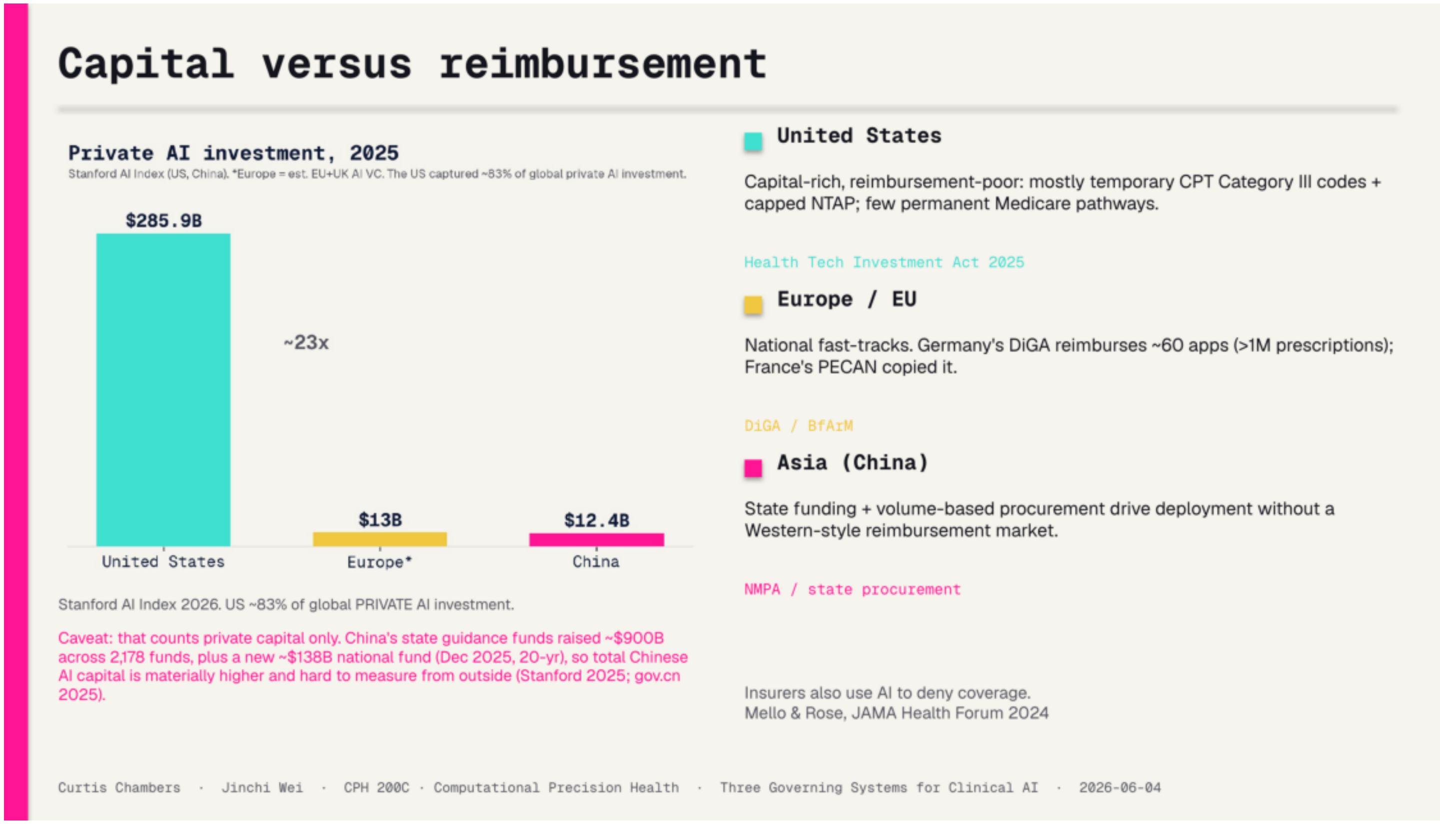
Slide 5 — Regulating clinical AI

Policy
Three regulatory philosophies, straight from the clinical literature. US clears the most devices (~1,451, 76% radiology, per Bedi et al. in npj Digital Medicine) device-by-device, but has no pathway for generative/LLM tools. Europe regulates hardest: AI Act high-risk stacked on the existing device law, a double conformity burden. China is state-coordinated, fewer approvals but mostly higher-risk Class III with trials (JMIR 2026). Footnote: Han et al. in Lancet Digital Health found clinical-AI RCTs are still few and early, so the evidence base is thinner than the device counts suggest. Aside if asked about general US AI policy: a June 2026 EO added a voluntary pre-release frontier-model review and a Treasury cyber clearinghouse, but it is non-clinical and bars mandatory licensing, so it does not change the clinical picture. Transition: who pays.



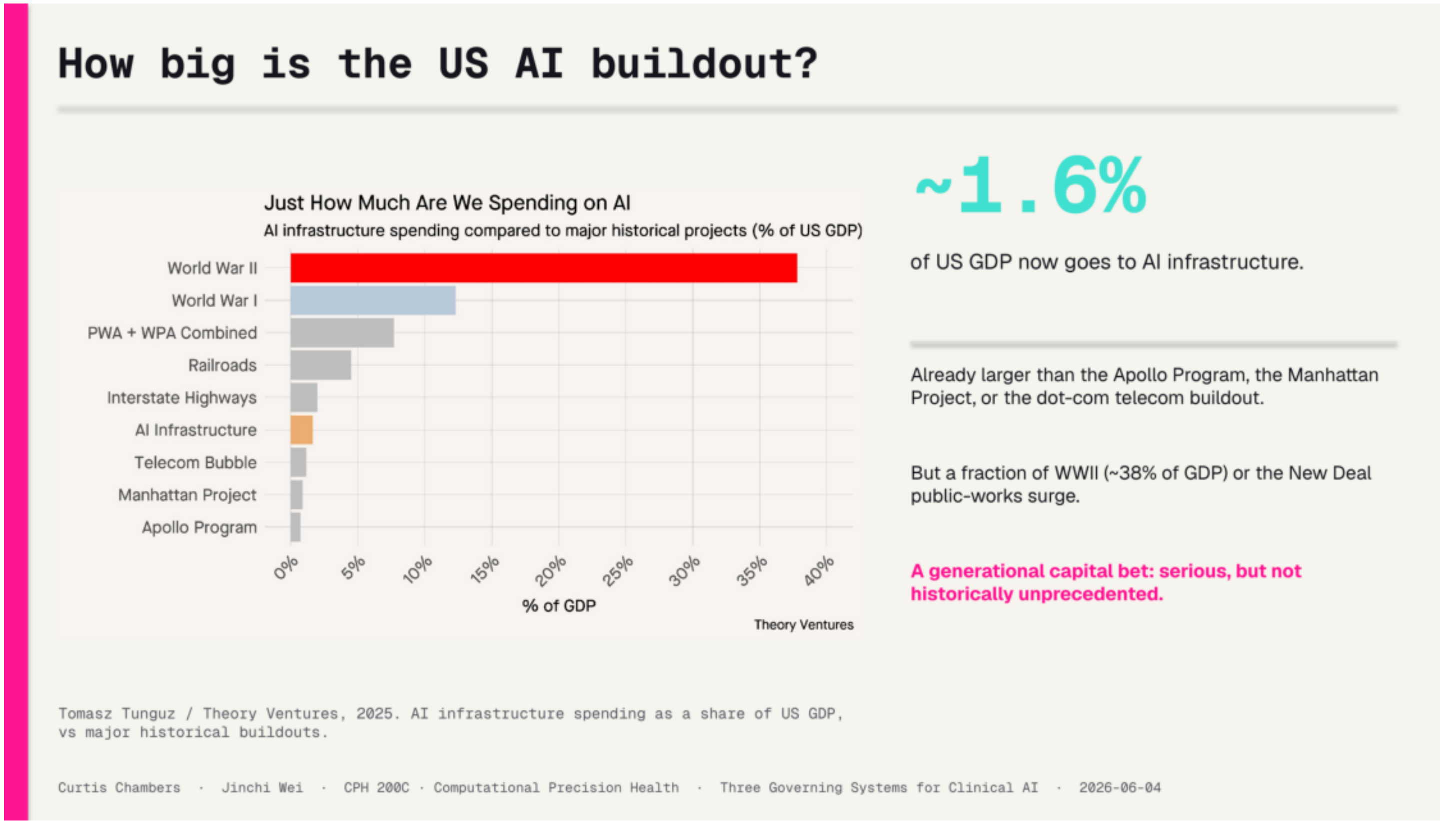
Slide 6 — Funding

Transition: capital versus the reimbursement pathway.



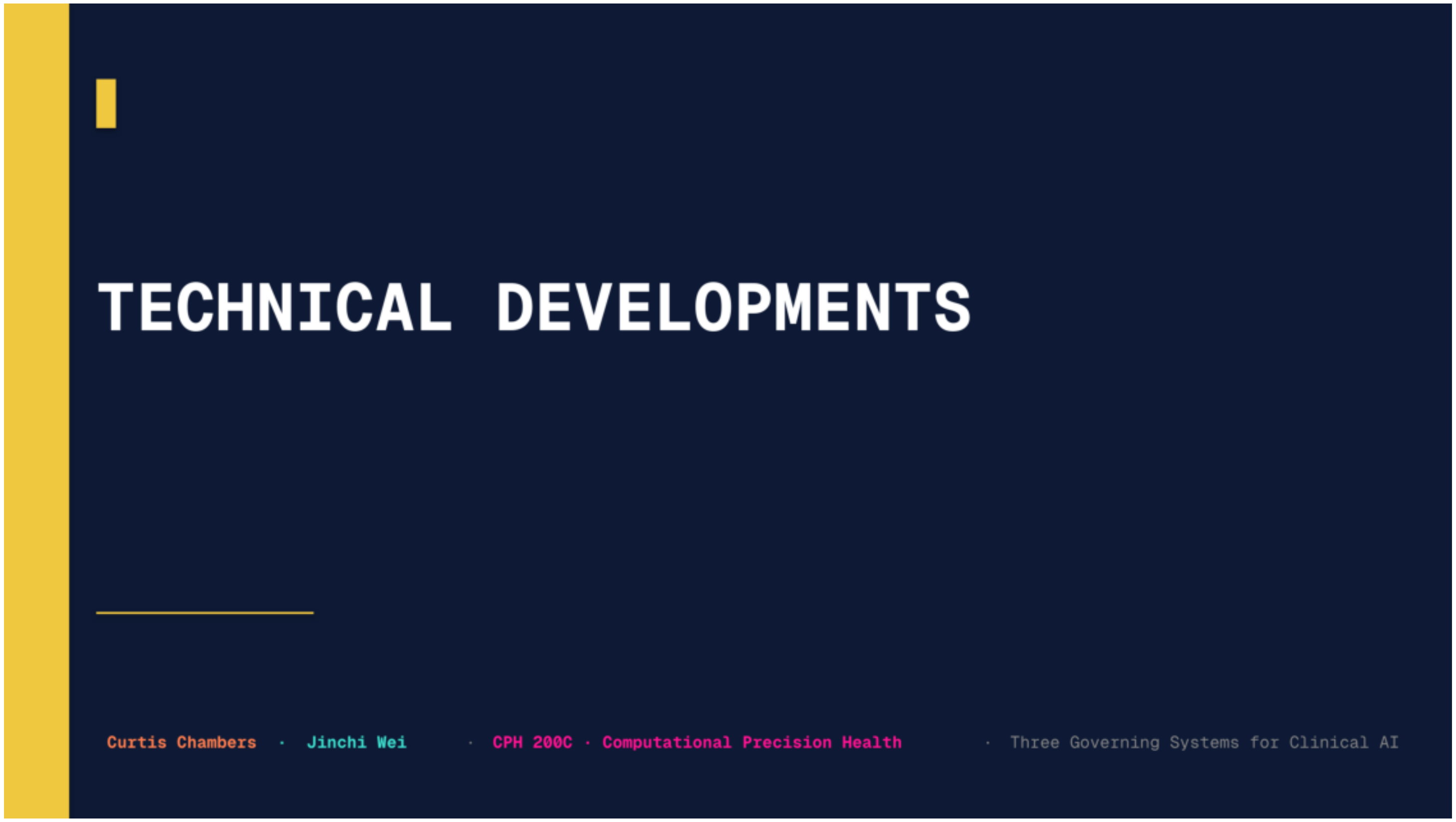
Slide 7 — Capital versus reimbursement

Funding
The twist for clinical AI: the US has ~83% of global private AI investment (Stanford AI Index), but capital is not what gets a tool used, reimbursement is, and there the US is poor (temporary CPT codes, capped add-on payments). Europe leans on national fast-tracks, Germany's DiGA actually works (~60 apps, >1M prescriptions). China sidesteps reimbursement entirely via state procurement. Caveat under the chart: that 83% is PRIVATE investment only, so it undercounts China, whose state guidance funds raised roughly 900 billion dollars across 2,178 funds, plus a new 138-billion-dollar national fund in December 2025, meaning total Chinese AI capital is materially higher and hard to measure from outside. For clinical AI, though, the payment pathway is still the binding form of funding. Transition: and just how big is the US bet?



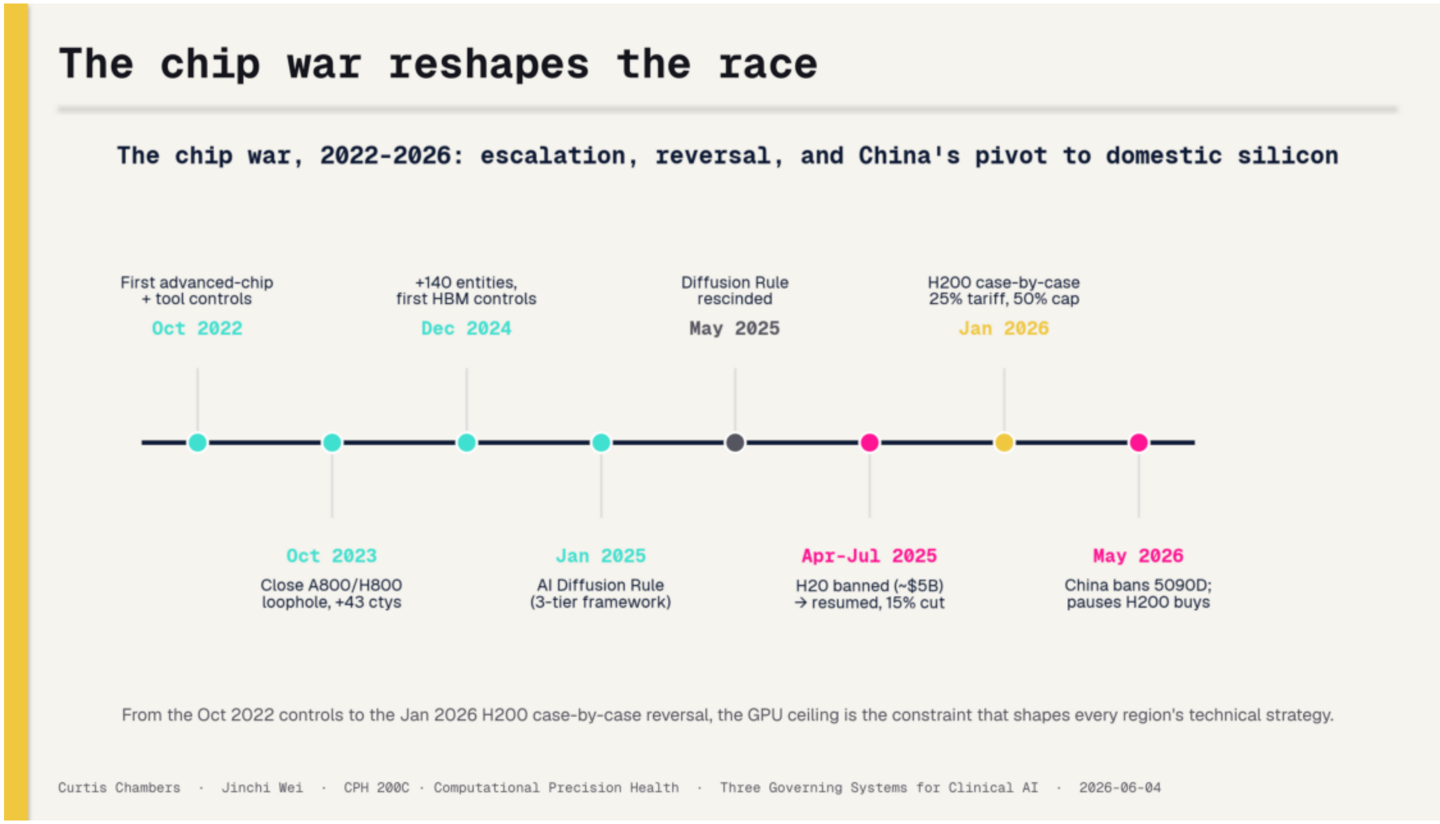
Slide 8 – How big is the US AI buildout?

US AI buildout, in scale
A quick scale check, with a clean third-party chart from Tomasz Tunguz of Theory Ventures. It plots AI infrastructure spending as a share of US GDP against the country's great buildouts. AI infrastructure is about 1.6 percent of GDP, already larger than the Apollo Program, the Manhattan Project, or the dot-com telecom buildout, but a fraction of World War II at roughly 38 percent or the New Deal public-works era. The funding takeaway: the US AI capital story is a serious, generational bet, but it is not historically unprecedented, which is worth holding in mind when we ask whether the spend pays off. Transition: that money still has to clear the technical constraint.



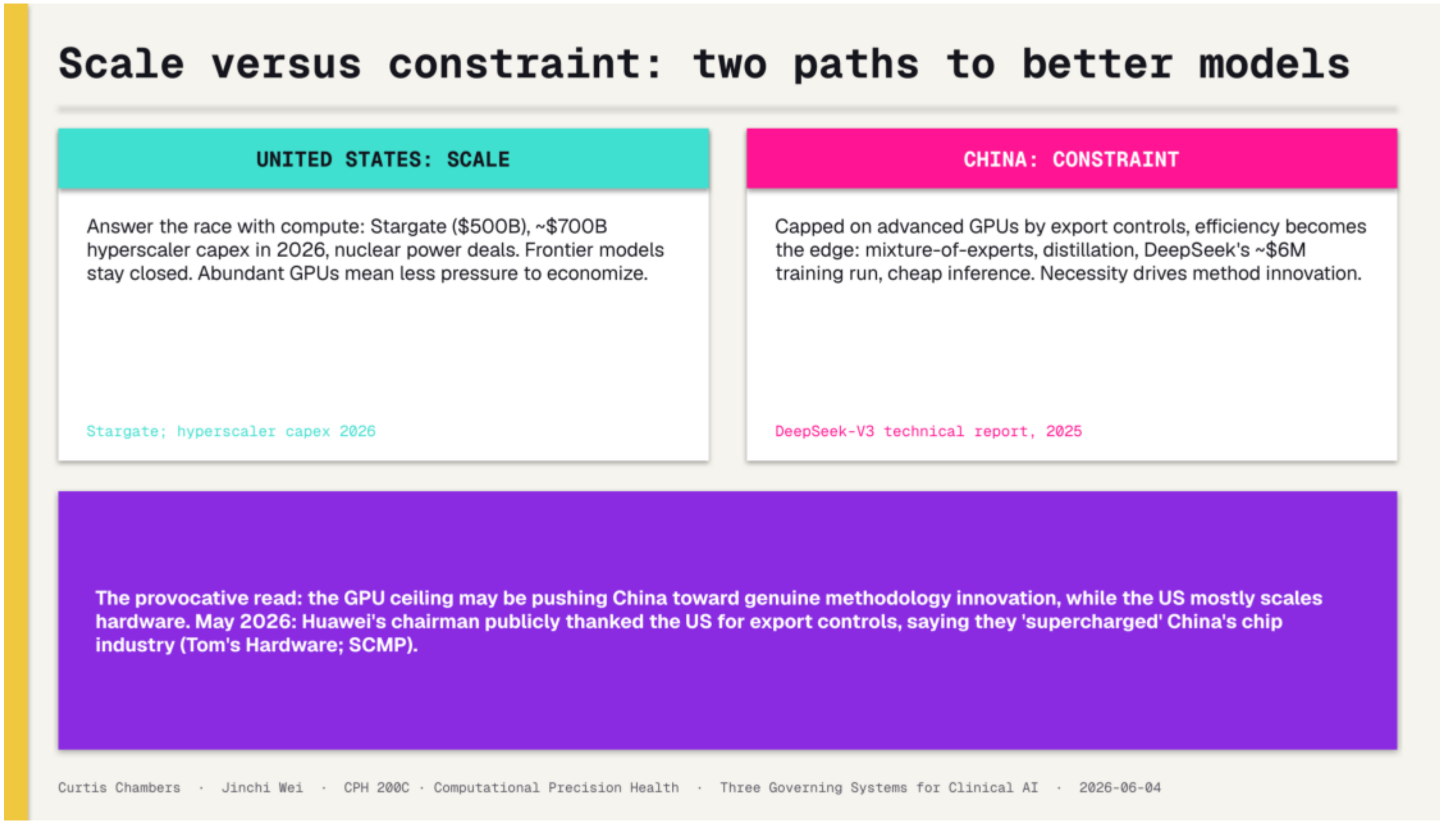
Slide 9 — Technical developments

Transition: the chip constraint and the two innovation paths it forces.



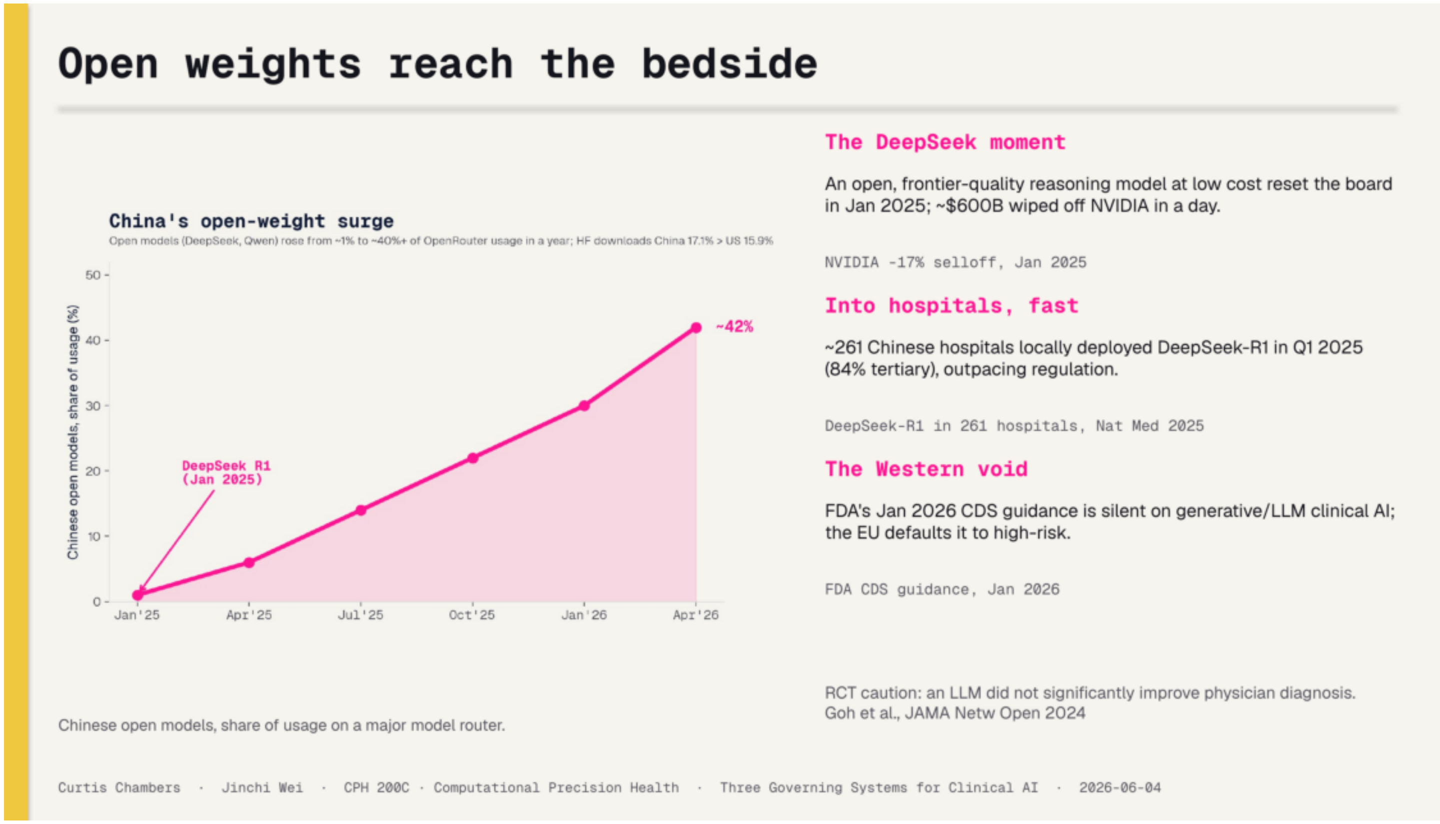
Slide 10 — The chip war reshapes the race

Chip war
How to read it: a timeline of US export controls and the NVIDIA-China saga. What to say: from October 2022, the US progressively tightened controls on advanced AI chips, closing loopholes, adding the H20 ban, then abruptly reversing in 2025-26 to case-by-case H200 sales with a tariff. The single point: the GPU ceiling is the binding constraint that shapes how every region pursues better models, which sets up the next slide. Transition: it pushed two very different innovation paths.



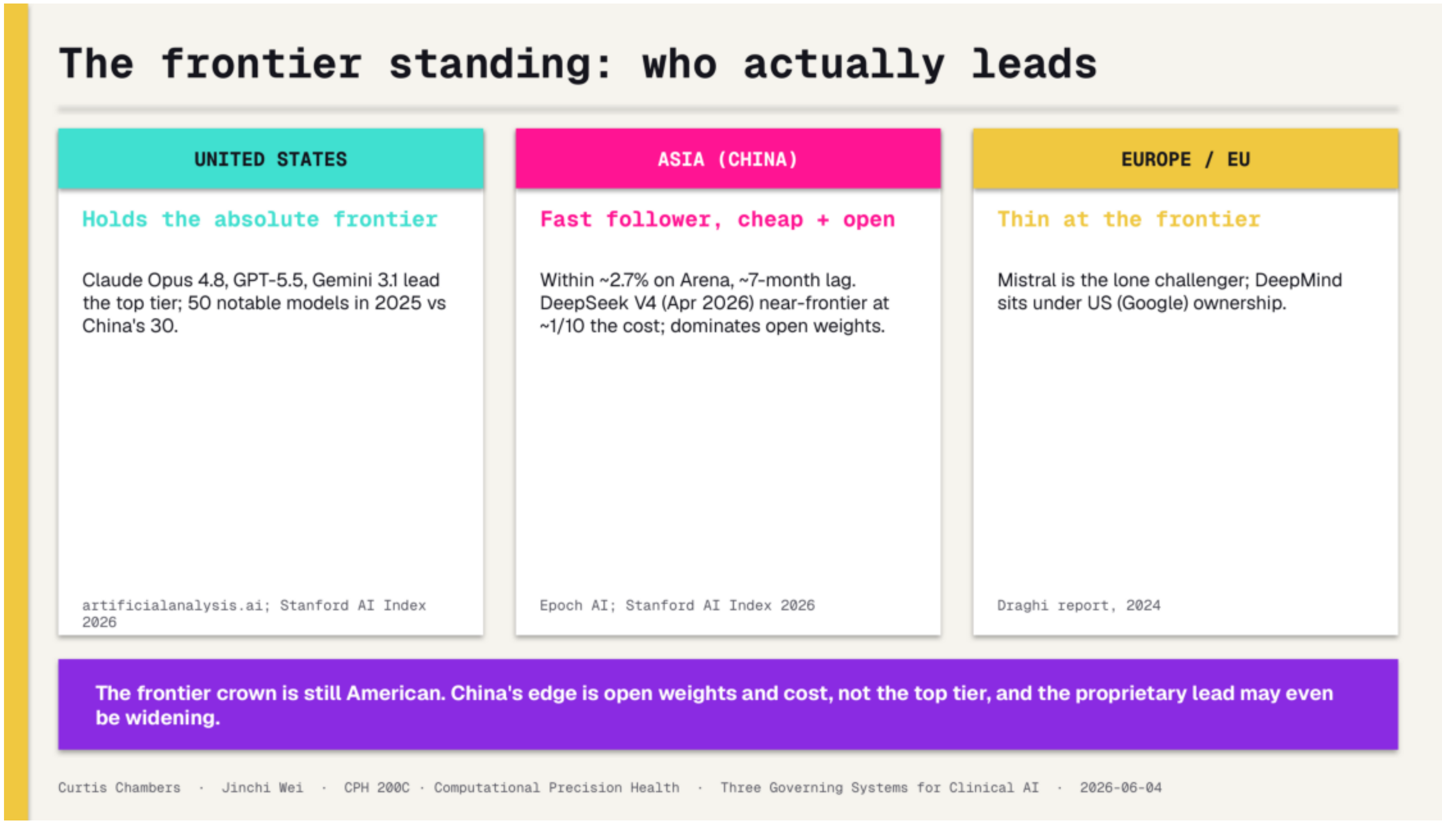
Slide 11 — Scale versus constraint: two paths to better models

Scale vs constraint
This is the most interesting technical point, and it represents both the US and China. The US answers the compute race with scale, Stargate's 500 billion dollars, roughly 700 billion in 2026 hyperscaler capex, nuclear deals, and keeps frontier models closed; abundant GPUs mean little pressure to economize. China, capped on advanced chips by export controls, is pushed the other way, toward method innovation: mixture-of-experts, distillation, DeepSeek's roughly 6-million-dollar training run, efficient inference. This slide is deliberately US vs China, the two sides actually racing on the model frontier. The provocative read, worth saying out loud: the GPU ceiling may be backfiring, driving genuine methodology innovation in China while the US mostly throws more hardware at the problem. Fresh evidence, end of May 2026: Huawei's rotating chairman publicly thanked the US for the export controls, saying they supercharged China's chip industry, and Huawei pitched a data-throughput scaling law to route around the EUV lithography it cannot buy. Treat the architecture claim as unverified, SCMP itself asks breakthrough or hype, but the posture, China openly framing the controls as counterproductive, is the point. Transition: and that open, efficient Chinese model is exactly what reached hospitals.



Slide 12 — Open weights reach the bedside

Technical
The open-vs-closed split reaches medicine fastest where oversight is state-coordinated. After DeepSeek's open reasoning model landed (Jan 2025), a Nature Medicine study documented ~261 Chinese hospitals locally deploying it within one quarter, mostly tertiary centers, deployment outran regulation. Contrast the West: FDA's Jan 2026 CDS guidance is silent on generative AI; the EU defaults LLM clinical tools to high-risk. The model exists everywhere; one system let it reach the bedside at scale. Transition: who deploys, overall.

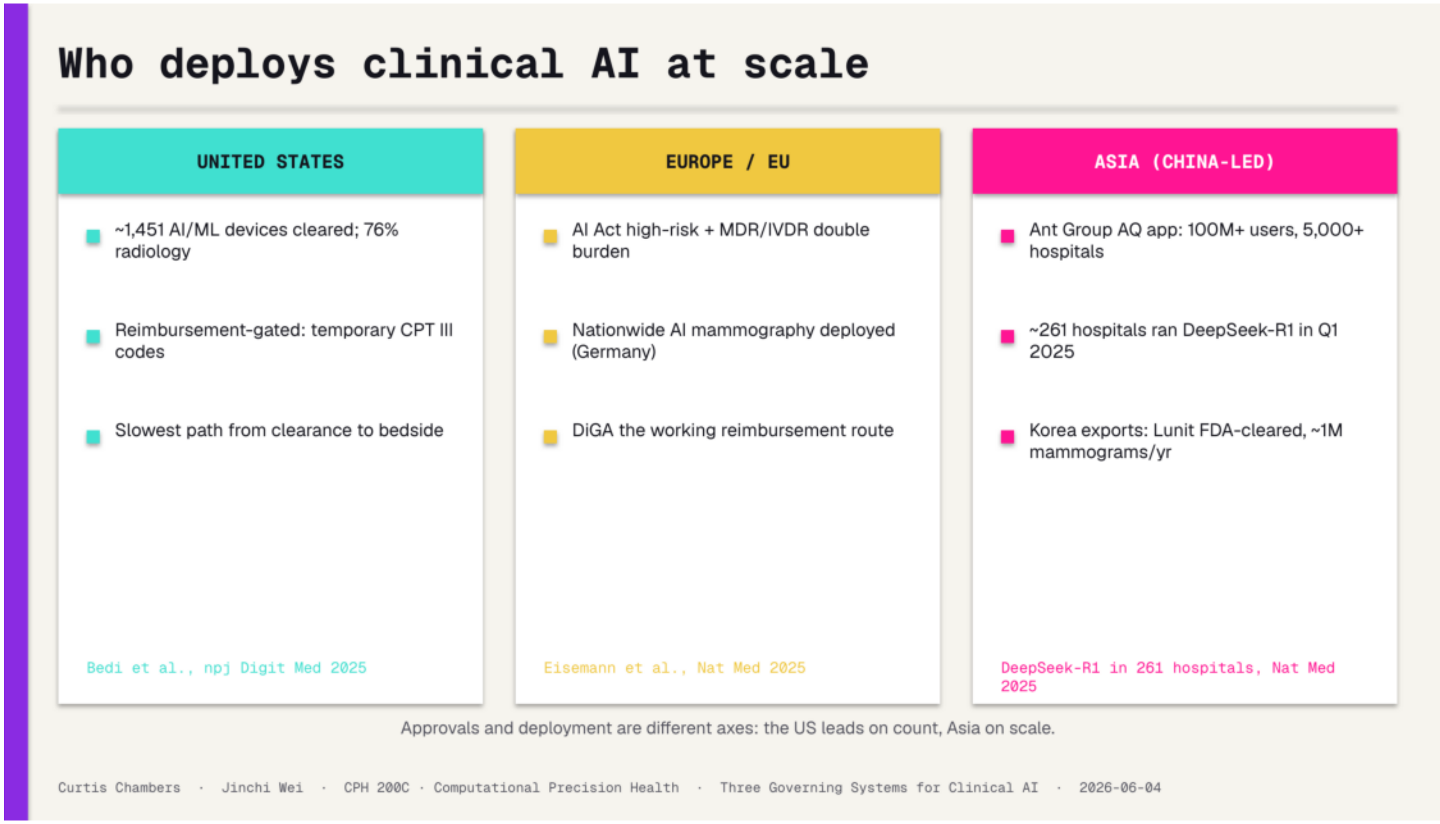


Slide 13 — The frontier standing: who actually leads

Frontier standing
The honest balance, and the slide that gives the US its due. For all of China's open-weight momentum, the absolute frontier as of mid-2026 is still American: Claude Opus 4.8, GPT-5.5, and Gemini 3.1 lead the top tier, and the US produced 50 notable models in 2025 to China's 30. China is a fast follower, within about 2.7 percent on the Arena leaderboard and lagging roughly 7 months on average (Epoch AI), and DeepSeek V4 in April 2026 reached near-frontier quality at about a tenth of the cost, which is itself the constraint-breeds-efficiency story. But near-parity-and-cheaper is not the same as leading. The band says it: the frontier crown is still American, China's edge is open weights and cost, and the proprietary top-tier lead may even be widening. Transition: so how do we even measure these clinical models?

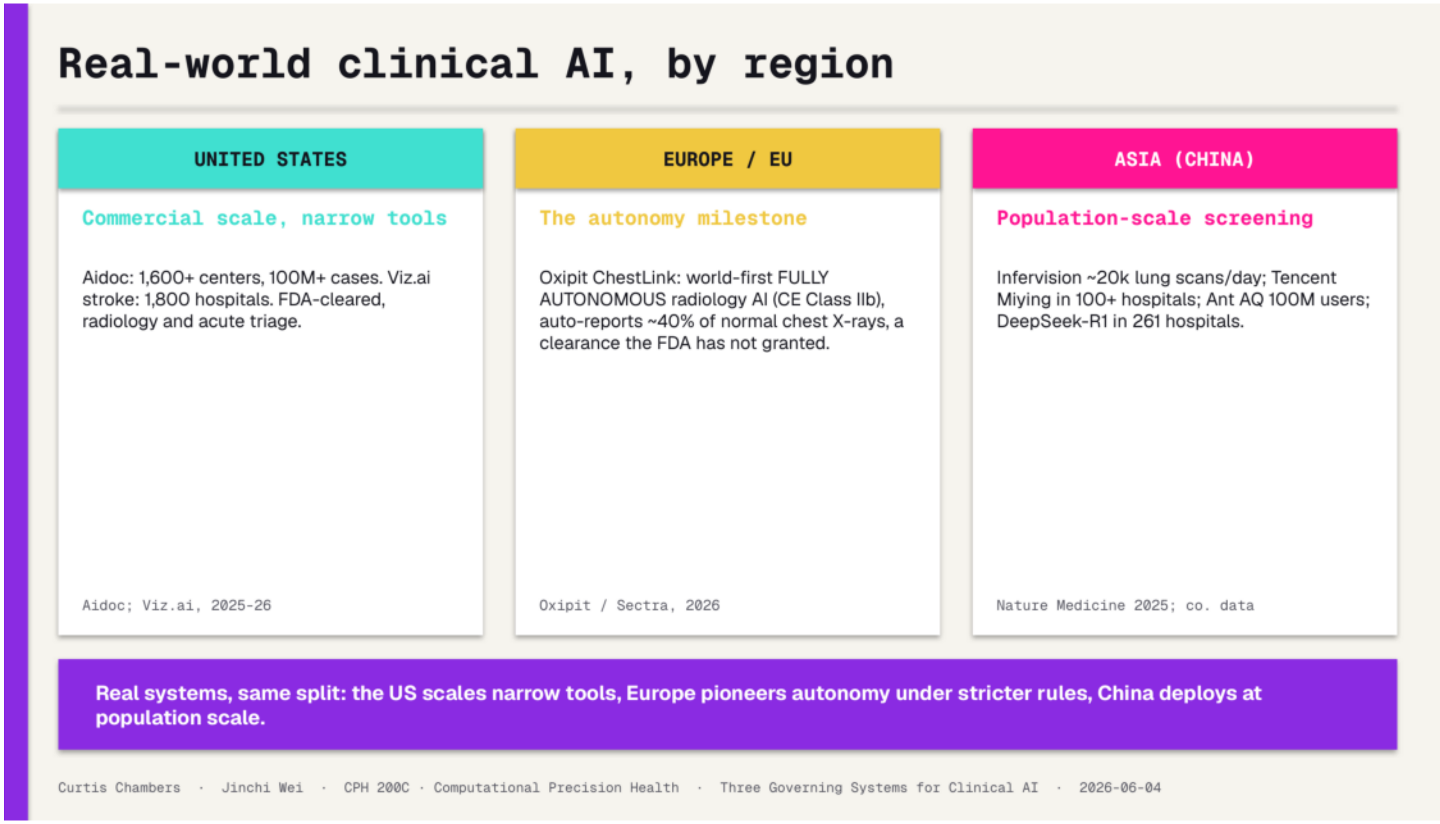
Slide 14 — Landscape

Transition: who actually deploys, and whether it works.



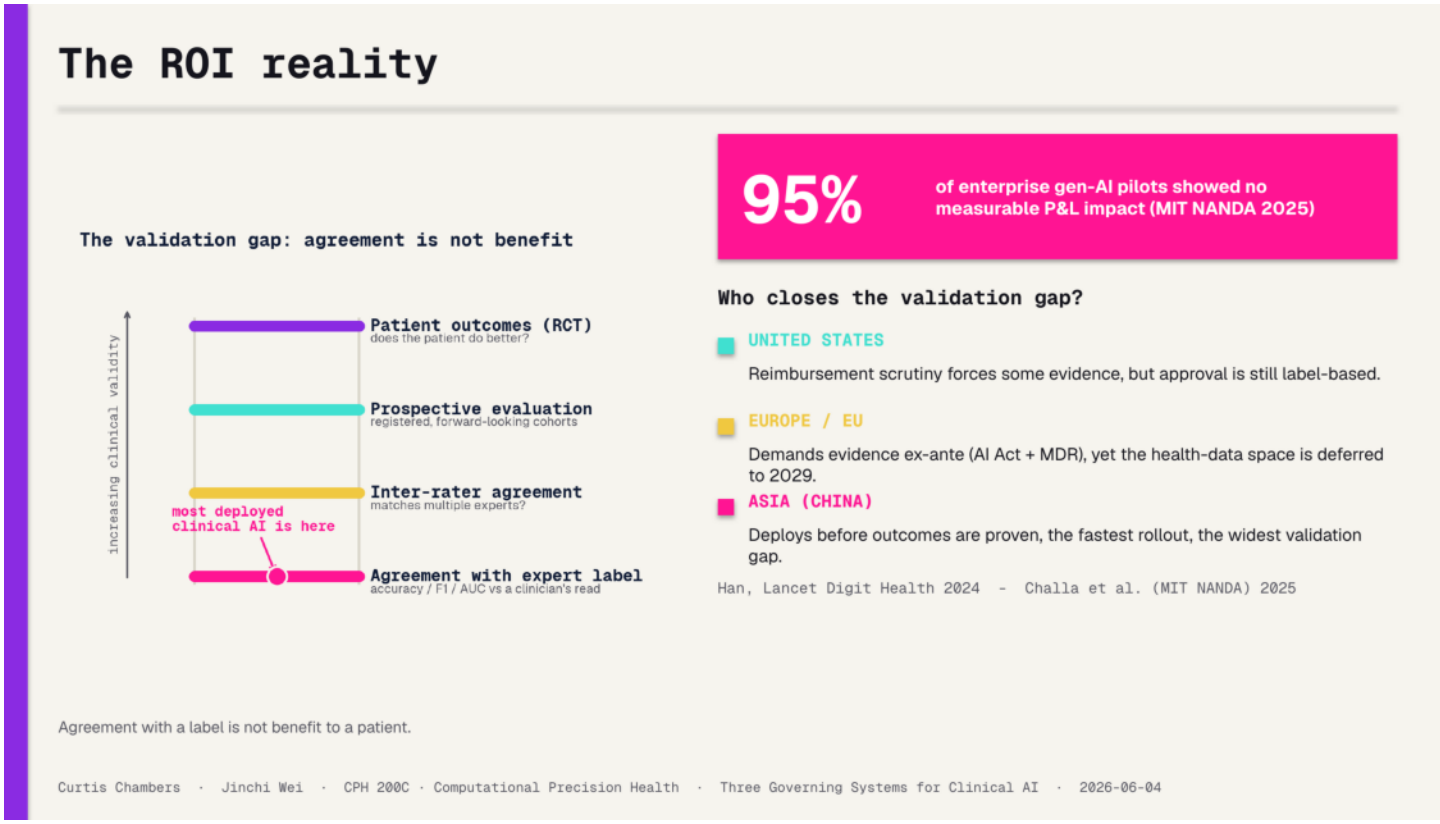
Slide 15 — Who deploys clinical AI at scale

Landscape: who deploys
Deployment inverts the approval count. The US approves the most devices but they are narrow and reimbursement-bottlenecked, so bedside reach is slow. Europe is slowest, gated by its own AI Act plus device law. Asia, led by China, deploys at a scale neither matches, Ant's AQ app reaches 100M+ users and 5,000+ hospitals, ~261 hospitals ran DeepSeek-R1 in Q1 2025 (Nature Medicine), and Korea exports FDA-cleared tools like Lunit.
Approvals and deployment are different axes. Transition: but does deployment at scale actually help patients?



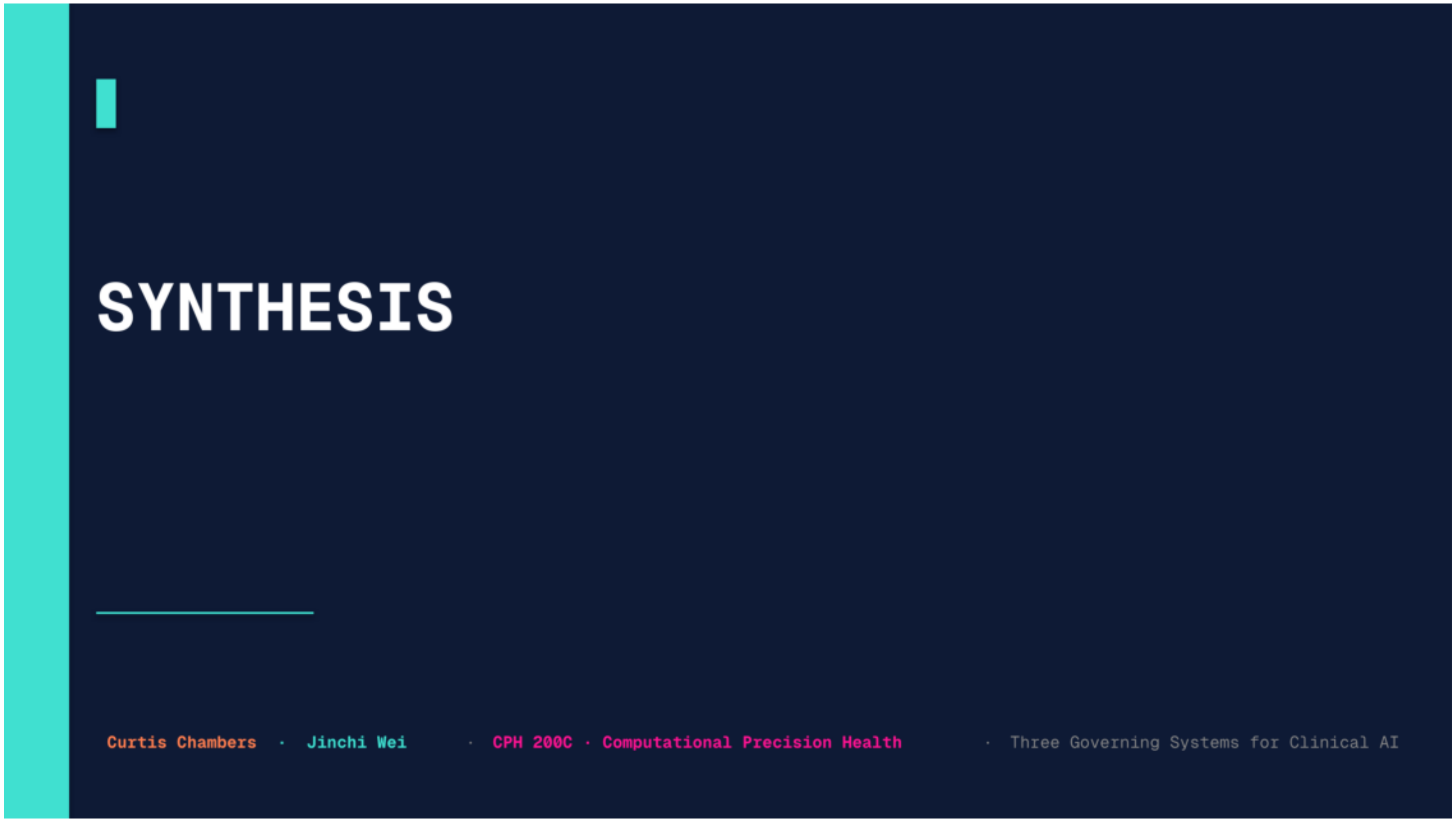
Slide 16 — Real-world clinical AI, by region

Real-world clinical AI
Concrete, named systems actually in hospitals, not abstractions. United States: the deepest commercial scale of narrow FDA-cleared tools, Aidoc is in over 1,600 medical centers with 100 million-plus cases analyzed, Viz.ai's stroke platform is in 1,800 hospitals. Europe: fewer deployments but the autonomy milestone, Oxipit's ChestLink is the world's first fully autonomous radiology AI, CE Class IIb, auto-reporting roughly 40 percent of normal chest X-rays with no radiologist, a level of autonomy the FDA has not cleared. China: population-scale screening, Infervision runs about 20,000 lung scans a day, Tencent Miying is in over 100 hospitals, Ant's AQ app reaches 100 million users, and DeepSeek-R1 went into 261 hospitals. The band: same regulate-build-deploy split, now in real systems, US scales narrow tools, Europe pioneers autonomy under stricter rules, China deploys at population scale. Transition: but does any of it pay off?



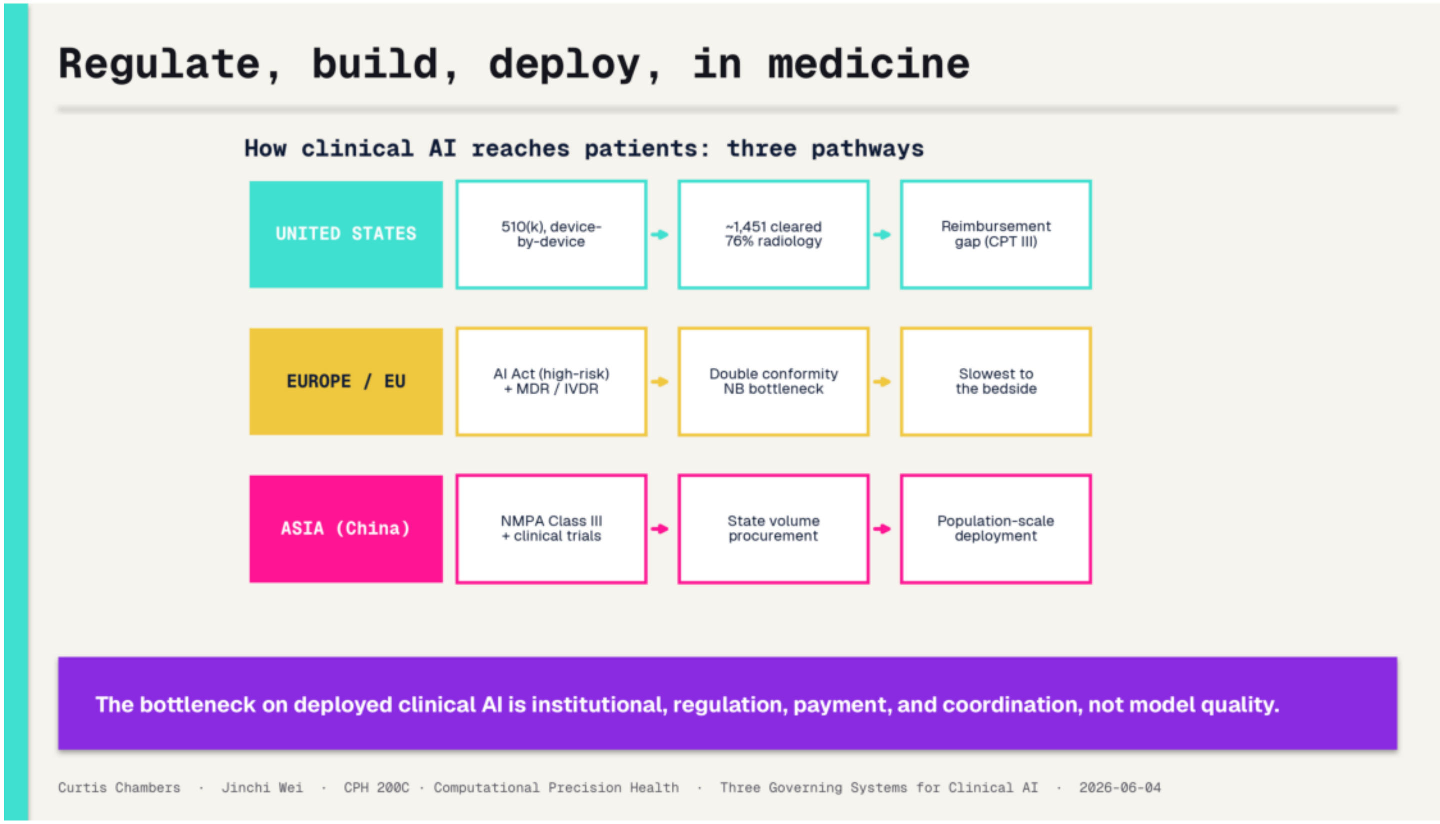
Slide 17 — The ROI reality

Landscape: ROI reality
The sober counterpoint, and the most research-grounded slide. MIT NANDA found 95% of enterprise gen-AI pilots delivered no measurable financial impact. And in clinical AI specifically (the ladder figure): most tools are validated against expert labels, not patient outcomes, so high agreement with a clinician is not the same as being right about the patient. Han et al. (Lancet Digital Health) make the same point, the RCT evidence is thin. That gap, between deployed and beneficial, is exactly the deployed-clinical-models problem. Transition: pull it together.



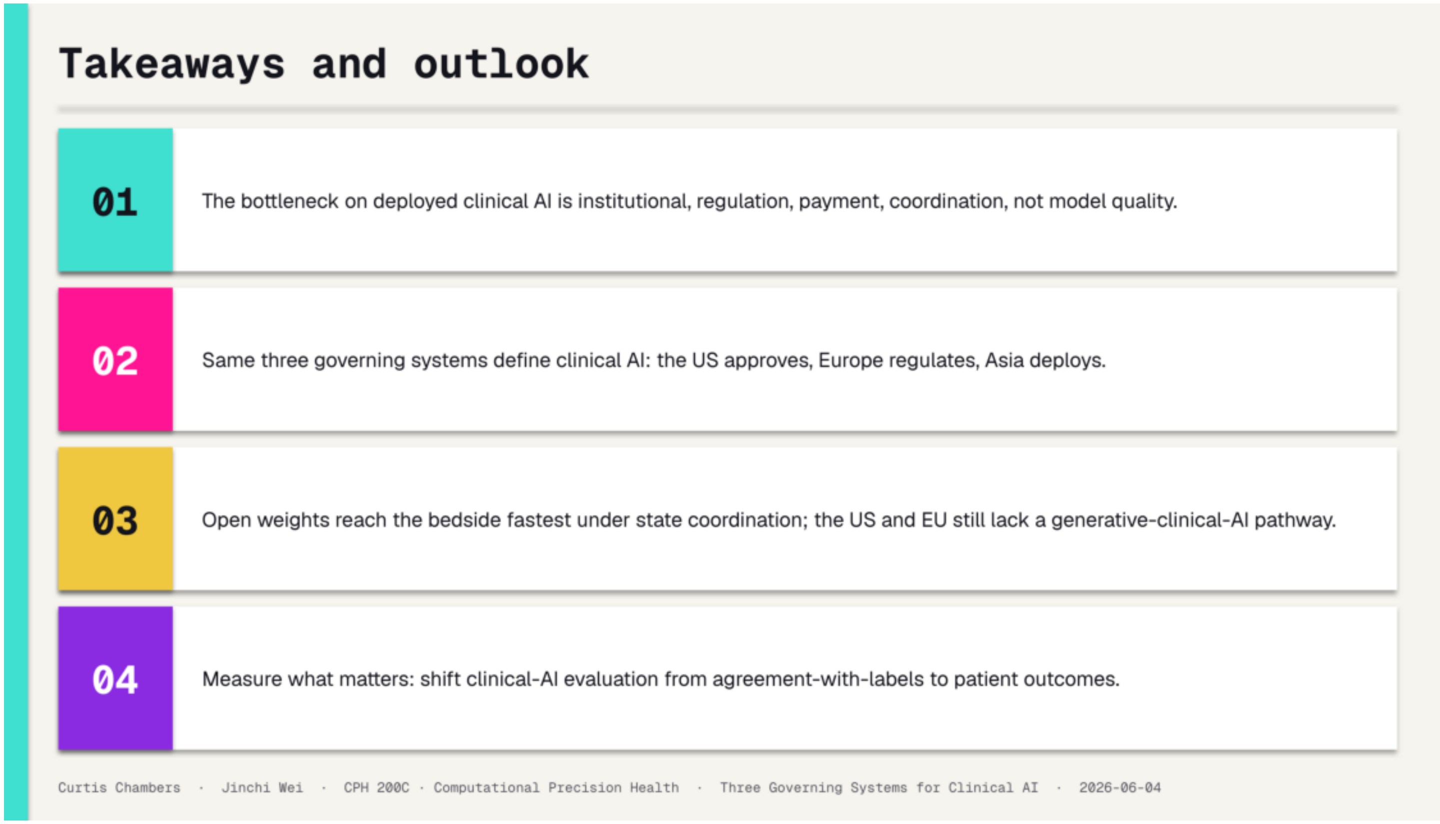
Slide 18 — Synthesis

Transition: the split in medicine, takeaways, references.



Slide 19 — Regulate, build, deploy, in medicine

Synthesis
The whole talk in one figure: three pathways clinical AI takes to patients. US approves device-by-device but stalls at reimbursement. Europe stacks the AI Act on device law, the slowest. China runs NMPA to state procurement to population-scale deployment. The band states the lesson: whether clinical AI reaches patients is decided by the institutional governing system, regulation, payment, coordination, more than by the model.
Transition: takeaways.



Slide 20 — Takeaways and outlook

Takeaways
Four closers: one, the bottleneck is institutional not technical. Two, the regulate-build-deploy split holds, US approves, Europe regulates, Asia deploys. Three, open weights reach the bedside fastest under state coordination, while the US and EU lack a generative-clinical pathway. Four, the field needs to measure patient outcomes, not label-agreement. Then references and questions.

Key references

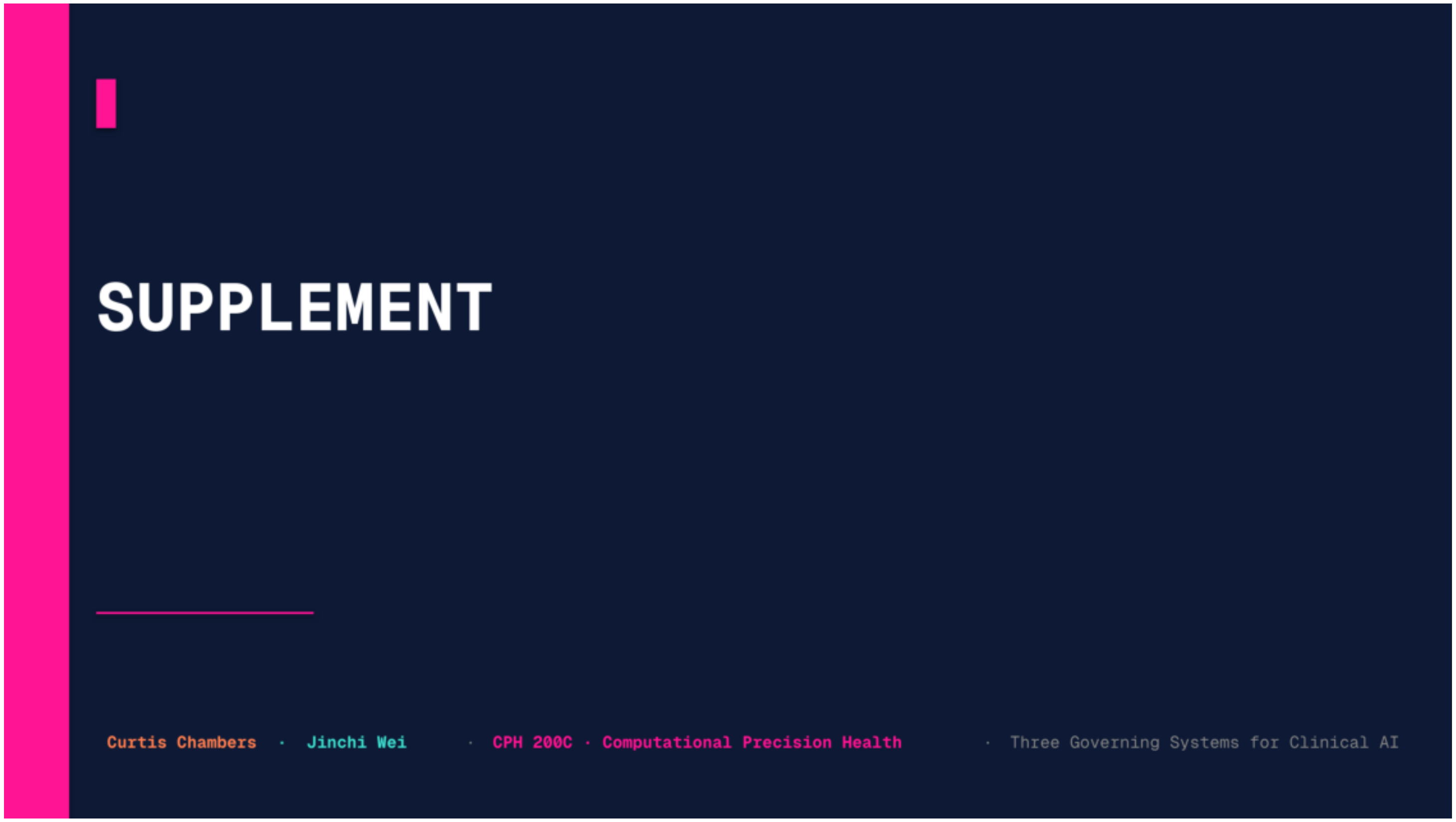
#	Reference	Venue / year
1	Han et al. - RCTs evaluating AI in clinical practice	Lancet Digital Health 2024
2	Goh et al. - LLM influence on diagnostic reasoning (RCT)	JAMA Network Open 2024
3	Eisemann et al. - Nationwide AI in mammography screening	Nature Medicine 2025
4	Bedi et al. - FDA AI/ML-enabled device landscape	npj Digital Medicine 2025
5	China NMPA AI-device approvals (review)	JMIR Med Informatics 2026
6	DeepSeek-R1 deployed across Chinese hospitals	Nature Medicine 2025
7	Mello & Rose - AI tools and insurance coverage denials	JAMA Health Forum 2024
8	Maslej et al. - Stanford AI Index Report 2026	Stanford HAI 2026
9	Challa et al. (MIT NANDA) - State of AI in Business	MIT NANDA 2025
10	Arora et al. - HealthBench	arXiv 2025
11	DeepSeek-AI - DeepSeek-V4 technical report	arXiv 2026
12	Epoch AI - US-China model capability gap	Epoch AI 2026

Framing: Lee, AI Superpowers (2018) - Schmidt et al., The Age of AI (2021) - Draghi, EU Competitiveness Report (2024).

Full source list with URLs in the companion research.md.

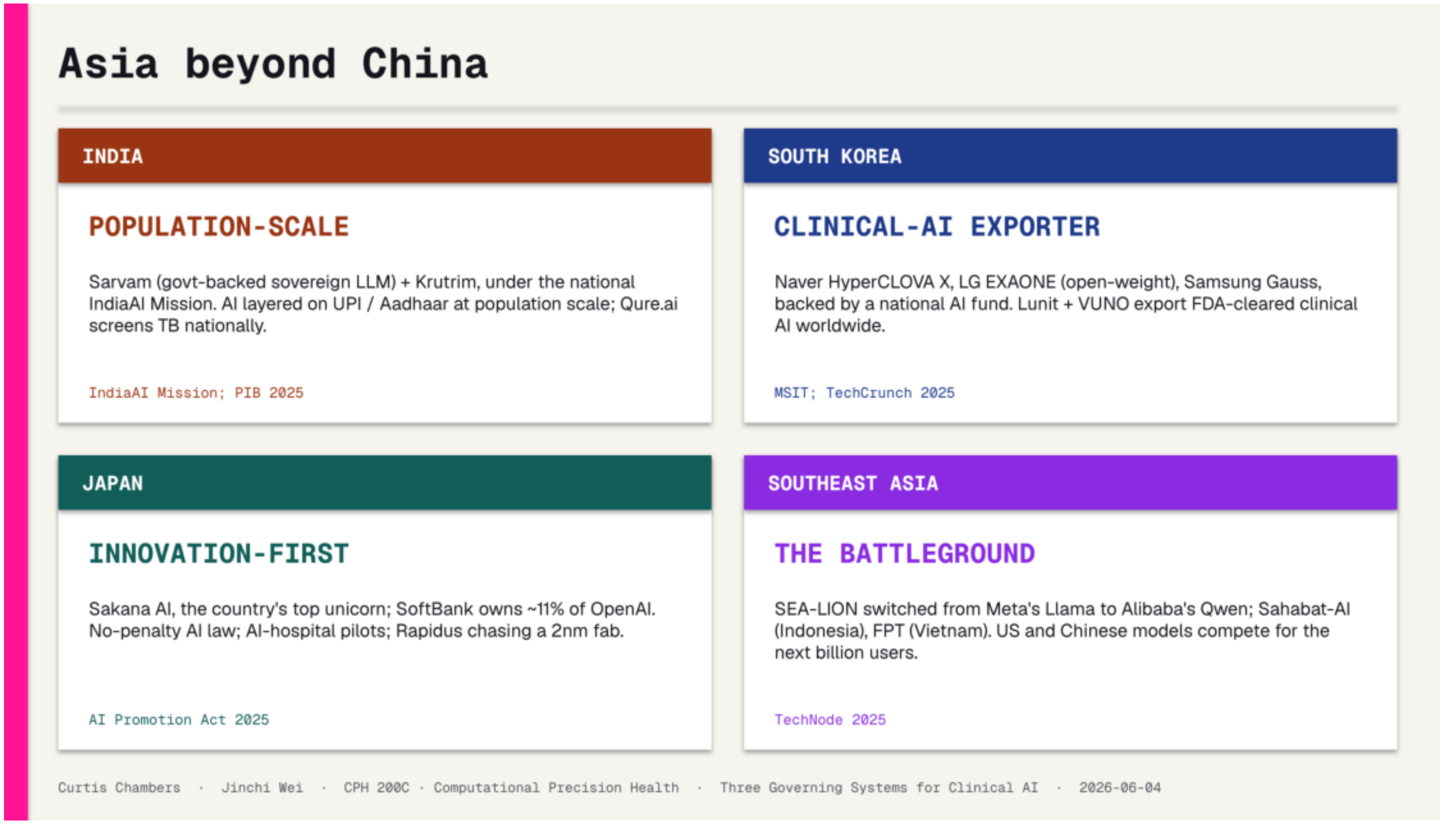
Slide 21 — Key references

References
Leave up for Q&A. The spine is peer-reviewed and recent: Han et al. (Lancet Digital Health) and Goh et al. (JAMA Network Open) on clinical-AI RCTs, Eisemann et al. (Nature Medicine) on deployed mammography AI, Bedi et al. (npj Digital Medicine) on the FDA device landscape, the Nature Medicine DeepSeek-in-hospitals study, plus the Stanford AI Index and MIT NANDA for the macro picture. Several are from the assignment's own reading list.



Slide 22 — Supplement

Backup material beyond the core talk: Asia beyond China.



Slide 23 — Asia beyond China (supplement)

Backup slide if time allows or if asked: the third governing system is not one country, so I frame each by its distinctive role rather than by spend. India is population-scale: a government-anointed sovereign model (Sarvam) and AI layered onto UPI and Aadhaar, with Qure.ai screening tuberculosis nationally. South Korea is the clinical-AI exporter: open-weight EXAONE and Naver at home, and Lunit and VUNO selling FDA-cleared clinical AI abroad. Japan is innovation-first, a no-penalty AI law, Sakana as its top unicorn, and SoftBank's roughly 11 percent stake in OpenAI. Southeast Asia is the live battleground, where its flagship SEA-LION model switched from Meta's Llama to Alibaba's Qwen. The takeaway: when we say Asia, China leads, but Korea, Japan, and India each run their own stack.


Thank You
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